times during a laser pulse exhibiting spikes.

Impurity-related fluctuations of ns breakdown in bulk transparent dielectrics have been reported in studies in which the laser beam was only weakly focused at *NA*s below 0.1 [109,200,201]. Under such circumstances, the probability of hitting upon particulate impurities is much larger than for the large *NA*s investigated in the present study because the focal volume V increases rapidly with decreasing *NA*: $V \propto 1/NA^4$. For sufficiently large *NA*, plasmas are compact [54,108,202] and likely smaller than the average distance between impurities. Therefore, impurities will hardly matter, and deterministic behavior can be expected not just for ultrashort laser pulses but also for ns breakdown by laser pulses with smooth temporal shape.

In our study, the influence of the temporal laser pulse shape is addressed by comparing the breakdown behavior of longitudinal single- and multimode laser pulses. Optimum reproducibility of all other irradiation parameters is ensured by using laser beams of high quality and diffraction limited focusing conditions. The potential influence of impurities is explored by varying the focusing strength between $0.3 < NA < 0.9$, corresponding to an 81-fold variation of the focal volume.

2.4. Scaling of laser-induced effects with pulse energy

Laser effects in transparent dielectrics may range from free-electron mediated chemical changes [22,198,203,204] through a phase transition up to vigorous shock wave emission and cavity formation [22,40,114,121,123,176,205]. In the present paper, we use the onset of a phase transition (bubble formation) in water as benchmark for the optical breakdown threshold and employ the maximum bubble radius R_{max} as convenient measure for the magnitude of the laser-produced effects above threshold [117]. This approach, which was introduced by Vogel et al. [117], facilitates a detailed exploration of scaling laws in a large (τ_L, λ_L, E_L) parameter space because the R_{max} (E_L) scaling can be fast and precisely determined through pump-probe measurements. Furthermore, the bubble radius provides a uniform measurement parameter over a very large range of plasma energy densities, whereas for a similar energy range in solids the damage type changes from colour center formation [206] through refractive index modifications [207] to cavity and crack formation [208,209].

2.5. Characterization of energy partitioning as a function of plasma energy density

The R_{max} (E_L) curves are a convenient measure for assessing the magnitude of laser-induced effects in water and soft tissue. However, additional information is required to fully characterize the hydrodynamic effects that besides bubble formation also include shock wave emission. The volumetric

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energy density $U_{\mathrm{Plasma}} = E_{\mathrm{abs}} / V_{\mathrm{Plasma}}$ of the laser-produced plasma is a key parameter governing the partitioning of absorbed laser energy in fractions going into vaporization, bubble formation, and shock wave emission.

The lower end of the plasma energy density scale in our investigations is given by the threshold for homogeneous bubble nucleation, which for ns pulses equals the kinetic spinodal limit and amounts to $U_{\mathrm{th}} \approx 1.2$ kJ cm^{-3} [188]. For laser pulse durations shorter than the acoustic transit time through the plasma, energy deposition is stress-confined and thermoelastic tensile stress reduces U_{th} to ≈ 0.6 kJ cm^{-3} [22,210]. Close to threshold, a large percentage of the absorbed energy goes into vaporization of liquid within the plasma volume, whereas in high-density plasmas most energy will be converted into the mechanical energy of shock wave and cavitation bubble produced upon plasma expansion [40,41,58]. The upper end of the plasma energy density in bulk dielectrics and the corresponding temperatures and pressures are of great interest to the scientific community [114,120,121,211,212] but there is still a need for systematic measurements of the energy density and partitioning covering a large range in (τ_{L}, λ_{L}) parameter space.

An experimental determination of U_{plasma} requires measurements of the absorbed energy fraction of the incident laser light, E_{abs}, and of the plasma volume, V_{plasma}. Although high-density ns plasmas emit bright radiation delineating the plasma shape [108], the luminescence from fs plasmas in bulk media is very weak, especially when the laser pulses are focused at small or moderate *NA*. That made it impossible to use plasma luminescence as indicator for fs breakdown and led researchers to introduce bubble formation as alternative criterion [110,113]. Thus far, data on the size of fs-laser produced plasmas rely on estimates [120,121,211,212], numerical modeling of the energy deposition [130,136,213-215], shadowgraphs of the plasma or the laser-induced phase transition [110,114,170,216-218], shadowgraphs combined with photoacoustic imaging [218,219], third harmonic imaging [220], or on the damage morphology in dielectric solids [208,221]. Evaluation of the damage morphology is useful for well-delineated low-density plasma effects but becomes problematic for high-density plasmas when cracks or cavities impair a precise localization of the plasma border. It is not at all possible for liquid breakdown because no permanent trace is left when the laser-induced bubble has vanished. A way for rapid in-situ determination of plasma size in water is offered by Favre's observation that white-light emission of fs plasmas becomes detectable when the laser pulses are focused at large *NA* [222]. In their study, water droplets were illuminated by a collimated laser beam, and light reflected at the droplet wall was focused back into the droplets. However, that goes along with strong spherical aberrations producing a caustic.

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In this paper, we take side-view photographs of the plasma luminescence produced by laser pulses focused through an aberration-free microscope objective. Recording of the plasma luminescence becomes possible by integration over the luminescence from a series of pulses.

Photographic determination of V_{plasma} is also a prerequisite for establishing an energy balance for the absorbed laser energy, E_{abs}. Knowledge of the plasma volume enables to determine the fraction of absorbed laser energy going into vaporization, E_{V}, when the luminescent region is completely vaporized. The bubble energy E_{B} is readily obtained from the maximum bubble radius (see section III.B). The absorbed energy can be determined by measuring the plasma transmission if most of the incident light is absorbed. This is approximately true for luminescent plasmas, where we found that backscattering and reflection amounts to less than 2% of the incident laser energy [223]. The shock wave energy can then be obtained by $E_{\text{SW}} = E_{\text{abs}} - E_{\text{V}} - E_{\text{B}}$ if other energy fractions play no significant role. Energy losses by plasma radiation leaving the luminescent region are neglected because very little energy leaves the water cell as plasma radiation. Literature values for blackbody temperatures of ns plasmas range from 10000 K for 4 mJ pulses [224] to 16000 K at much larger pulse energies ≥ 100 mJ [225,226]. Vogel et al. estimated the energy content of blackbody radiation from a luminescent ns plasma with $T_{\text{blackbody}}$ = 10000 K and found that it amounts to less than 0.1 % of the absorbed laser energy [114]. However, although little radiation energy leaves the plasma, we will see in section IV.C. that in nanosecond breakdown very hot regions are produced within the region of primary energy deposition, which leads to radiative redistribution of a significant energy amount *within* the luminescent region. Work done against surface tension and to overcome liquid viscosity must be considered for nano- and microbubble dynamics close to the breakdown threshold [42] but plays no role for millimeter-sized bubbles [63,66,227] and little role for the smallest bubbles produced by luminescent plasmas, which still have a size larger than 10 µm [42]. We showed previously, that neglecting this part of the energy balance as well as losses by plasma radiation, an indirect assessment of E_{SW} from E_{abs}, E_{v}, and E_{B} becomes possible, which avoids difficult measurements of shock wave amplitude and duration that would be needed for a direct determination of E_{SW} [40,114,228,229].

Knowledge of the plasma energy density provides also an independent starting point for determining plasma temperature, T_{plasma}, and pressure, p_{plasma}, different from spectroscopic approaches because these physical quantities are linked to mass density ρ_0 and energy density U (or internal energy, respectively) through the equation of state (EOS) of water [230]. The term "plasma temperature" here refers to the thermodynamic temperature after thermalization of electron energies, which is completed after 20-30 ps.

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